Topic: wearable head impact sensor high-g accelerometer drop test validation · Generated: 2026-11-11 23:30:28 UTC

12 papers compared · 12 full text · 0 abstract

The papers assembled here span a wide methodological spectrum—from MEMS fabrication and bench-top metrological characterization to machine learning classifiers trained on wearable IMU data—yet all share a concern with measuring or inferring human motion and impact through inertial sensing. The dominant independent-variable pairings involve sensor placement on the body, choice of classification algorithm, and signal preprocessing, with fall detection accuracy, recall, and F-score serving as the most frequently reported dependent variables; this pairing recurs across the wrist-based, waist-mounted, and multi-site body-placement studies. Several papers draw on overlapping or combinable public datasets—including UniMib, UP Fall, and UMA Fall—while others contribute newly collected samples from controlled participant cohorts, though population sizes remain modest and skew toward younger, healthy adults rather than the elderly populations most at risk. The smartwatch characterization study and the piezoresistive accelerometer fabrication paper stand apart from the fall-detection majority, addressing sensor-level accuracy and hardware sensitivity rather than downstream classification tasks, and neither reports drop-test or high-g validation protocols. Most conspicuously absent across the entire collection is any systematic high-g drop-test validation of the kind used to characterize head impact sensors: no paper reports calibrated free-fall or guided-drop experiments against a traceable reference at accelerations representative of head impacts, and the "high-g" regime is never explicitly targeted as an independent variable. The pedestrian bonnet study is the sole entry that addresses peak head acceleration and head injury criteria, yet it does so through multibody simulation and low-velocity dummy tests rather than instrumented wearable sensors, leaving the core topic of wearable head impact sensor validation under controlled high-g drop conditions essentially unaddressed by this body of work.

title	year	venue	Objectives	Problem Statement	Methods Used	Independent Variables	Dependent Variables	Dataset	Population Sample	Results	Findings	Conclusions	Contributions	Limitations	Challenges	Research Gap	Future Research	Practical Implications	Applications	Literature Survey	Summarized Abstract	Summarized Introduction	TL;DR
Intelligent Life Guard: An IoT-Enabled Smart Helmet for Vital Monitoring & Fall Detection ↗ jddt1635s5 full text	2026	International Journal of Drug Delivery Technology	- The objective of the Intelliguard Helmet is to enhance safety and health tracking in dangerous areas. - The helmet aims to provide real-time analytics and alerts to emergency personnel or supervisors.	The central research question is how to enhance safety and health tracking in dangerous areas using IoT technology.	- The helmet uses IoT theory to collect and share information through devices with sensors, actuators, and connectivity modules. - The system is set up with wokwi.com website and programming languages such as embedded C, ESP832, and proteins. - The helmet uses MQTT and HTTP protocols for communication with cloud platforms.	—	- The dependent variables include the wearer's condition, such as temperature and heart rate, and the environment, such as gas levels and proximity to hazards.	—	—	- The helmet can detect impacts, falls, or collisions and immediately alert supervisors and teams involved in rescue. - The helmet provides real-time information to be used in proactive safety measures. - The helmet can track the condition of the wearer and the environment in real-time.	- The helmet can detect impacts, falls, or collisions and immediately alert supervisors and teams involved in rescue. - The helmet provides real-time information to be used in proactive safety measures. - The helmet can track the condition of the wearer and the environment in real-time.	- The Intelliguard Helmet is an innovative solution to enhance safety and health in risky working conditions. - It allows real-time detection of the state of the wearer and the immediate environment. - The helmet demonstrates the power of IoT in improved safety, effectiveness, and decision-making.	- The Intelliguard Helmet is an innovative solution to enhance safety and health in risky working conditions. - The helmet provides real-time detection of the state of the wearer and the immediate environment. - The helmet demonstrates the power of IoT in improved safety, effectiveness, and decision-making.	- The Intelliguard helmet system may be pricier compared to conventional helmets, including initial purchase costs, maintenance, and subscription fees for cloud services.	- The challenges include providing real-time detection of the state of the wearer and the immediate environment. - The challenges also include enabling real-time monitoring of locations and instantaneous response to emergencies.	- The research gap is the need for innovative solutions to enhance safety and health in risky working conditions. - The gap is addressed by the Intelliguard Helmet, which provides real-time detection of the state of the wearer and the immediate environment.	—	- The Intelliguard Helmet can be applied to industries such as construction, mining, and sports, where employees and athletes are exposed to physical and environmental hazards. - The helmet enables real-time monitoring of locations and instantaneous response to emergencies. - The helmet provides hands-free control and displays essential information to the wearer.	- The applications include industries such as construction, mining, and sports, where employees and athletes are exposed to physical and environmental hazards. - The applications also include areas such as factories and cyclists.	- The literature review examines core theories relevant to smart helmets, including safety engineering principles, human computer interaction, IoT frameworks, and artificial intelligence of wearable technology. - Prior work by Shi.et.al and Kawale S. R Mallikarjun S. Gowda highlights the importance of real-time digital representation of helmet status.	- The Intelliguard Helmet is a combination of IoT technology to enhance safety, health tracking, and communication in dangerous areas. - The helmet has sensors to track impacts, fast motions, and physiological data in real-time. - The helmet provides hands-free voice communication and real-time analytics to avoid accidents and long-term health problems. - The Intelliguard Helmet can be equipped with features such as smart lighting and diagnostic alarms.	- The Intelliguard Helmet is a new product that utilizes the power of IoT to provide more safety to workers and individuals in dangerous environments. - The helmet is equipped with advanced IoT sensors to monitor physical conditions and surroundings of the wearer in real-time. - The helmet is designed to provide active protection by using intelligent sensors. - The helmet relays information to a central cloud-based system for analysis and real-time warnings.	The Intelliguard Helmet is an IoT-enabled smart helmet for vital monitoring and fall detection, using sensors and wireless communication to track the wearer's condition and environment in real-time. The helmet provides real-time analytics and alerts to emergency personnel or supervisors. The Intelliguard Helmet aims to enhance safety and health tracking in dangerous areas.
Simultaneous Bench-Based Metrological Characterization of Smartwatches' Accelerometers for Accurate Measurement ↗ es14060356 full text	2026	Technologies	- To propose a methodology for the simultaneous metrological characterization of multiple smartwatch accelerometers. - To evaluate the measurement discrepancies associated with the accelerometer integrated into a commercially available smartwatch. - To assess the accuracy of smartwatch measurements using a reference accelerometer.	The measurement reliability of accelerometers in consumer-grade smartwatches is often compromised, necessitating proper metrological characterization to ensure precision and consistency.	- A seismic table to generate controlled vibrations. - A reference accelerometer to assess the accuracy of smartwatch measurements. - A custom Wear OS application to collect triaxial accelerometer signals. - The Android SensorManager API to access raw accelerometer data streams.	- Frequency range of 1–8 Hz, - Acceleration amplitudes between 1 and 4 m/s2, - Sampling rate of 50 Hz	- Acceleration measurements, - Frequency response, - Amplitude accuracy	—	- Five consumer-grade smartwatch units, - Running Wear OS version 4.x or higher	- Five commercial smartwatch units were tested, - Collecting acceleration data at sampling rate of 50 Hz, - Acceleration amplitudes between 1 and 4 m/s2, - Frequency range of 1–8 Hz, - 15 amplitude and frequency combinations	- Variations in frequency response and amplitude accuracy across different smartwatch units, - Errors and uncertainties quantified following ISO standards, - The proposed methodology allows for consistent comparison of measurement performance across devices, - Reduction of the time required for the process	- The proposed simultaneous metrological characterization methodology is a viable and efficient approach, - The capability to calibrate multiple devices concurrently offers considerable advantages, - The proposed methodology allows for the simultaneous evaluation of all smartwatch accelerometers	- A methodology for simultaneous metrological characterization of multiple smartwatch accelerometers. - A approach for efficient and consistent bench-based measurement evaluation. - A method for assessing the accuracy of smartwatch measurements using a reference accelerometer.	- No stated limitations in the text	- Ensuring that excitation occurs along only one axis. - Isolating the contribution from the excitation produced by the seismic table or the effect of gravity. - Developing a methodology for simultaneous metrological characterization of multiple smartwatch accelerometers.	- The lack of proper metrological characterization of smartwatch accelerometers. - The need for efficient and consistent bench-based measurement evaluation. - The requirement for accurate assessment of smartwatch measurements.	- No specific future research directions stated in the text, - Potential for large-scale studies in healthcare research	- Efficient and consistent bench-based measurement evaluation, - Reduction of overall characterization time, - Improvement of efficiency in large-scale studies	- Health-related research applications, - Wearable movement monitoring, - Healthcare research	- The paper mentions ISO 16063-21:2003 and ISO 266:1997 as relevant standards. - The study builds upon prior work in wearable technology and healthcare. - The paper does not provide an extensive literature survey.	- Accelerometers in consumer-grade smartwatches have potential for health-related research applications but their measurement reliability is often compromised. - Proper metrological characterization is necessary to ensure precision and consistency. - This study proposes a methodology for simultaneous metrological characterization of multiple smartwatch accelerometers. - The methodology enables efficient and consistent bench-based measurement evaluation. - The results demonstrate variations in frequency response and amplitude accuracy across different smartwatch units.	- Wearable technology has introduced significant innovations in healthcare. - Wearable devices contribute to the delivery of more effective, personalized, and patient-centered care. - Recent advances in telecommunications and technology have allowed wearable devices to address and mitigate the limitations of traditional clinical practices. - The study focuses on the metrological characterization of smartwatch accelerometers.	This study proposes a methodology for simultaneous metrological characterization of smartwatch accelerometers, enabling efficient and consistent bench-based measurement evaluation. The methodology employs a seismic table to generate controlled vibrations and assesses the accuracy of smartwatch measurements. The results demonstrate the viability of this approach for characterizing acceleration measurements of consumer-grade smartwatches.
Contact-Triggered Active Bonnet System for Pedestrian Head Injury Mitigation: Multibody Modelling and Experimental Validation ↗ s26154743 full text	2026	Sensors	- To analyse pedestrian impact dynamics and evaluate the design performance of a contact-triggered active bonnet system. - To experimentally validate the model using controlled pedestrian impact tests. - To evaluate the influence of impact configuration and pedestrian trajectory on head injury severity.	Pedestrian head injuries represent one of the most severe consequences of vehicle-pedestrian collisions, primarily resulting from the impact between the pedestrian head and the vehicle hood or windshield region.	- Multibody modelling using MATLAB/Simulink and Simscape Multibody. - Experimental validation using controlled pedestrian impact tests. - Parametric sensitivity analysis to evaluate the influence of sensing and actuation parameters on pedestrian head injury mitigation.	- actuator torque, - vehicle impact velocity, - pedestrian positioning and head trajectory	- peak head acceleration, - HIC15 value, - head injury severity	- simulated: MATLAB/Simulink with Simscape Multibody environment, - experimental validation using controlled pedestrian impact tests	- a custom-built research dummy designed to reproduce the global kinematic behaviour of a pedestrian during vehicle impact	- peak head acceleration of 84 m/s^2, - HIC15 value of 264, - peak acceleration reduced to 76 m/s^2, - HIC15 value close to the baseline configuration, - actuator torque of 500 N•m, - actuator torque of 1000 N•m, - actuator torque of 1500 N•m	- the active bonnet system can reduce peak head acceleration and HIC15 values under several investigated impact conditions, - the protective effect is influenced by pedestrian positioning and head trajectory relative to the vehicle front-end geometry, - higher effectiveness when the head impact remains within the deformable bonnet region, - actuator torque represents a critical design parameter influencing the effectiveness of active pedestrian protection systems, - lower torque values generate a slower bonnet lifting motion, - higher actuator torques accelerate the bonnet deployment process	- the performance of the active bonnet system is strongly dependent on the synchronization between pedestrian detection and bonnet deployment, - optimizing both the sensing and actuation subsystems is crucial in real-world active bonnet implementations, - the active bonnet system can mitigate pedestrian head injury severity, - the system's effectiveness is influenced by impact configuration and pedestrian trajectory	- A contact-triggered active bonnet system for pedestrian head injury mitigation. - A multibody model of the vehicle-pedestrian system using MATLAB/ Simulink and Simscape Multibody. - Experimental validation of the model using controlled pedestrian impact tests.	- the study employs a simplified actuator representation, - the actuator model does not reproduce the detailed behaviour of pyrotechnic deployment systems, - the analysis is based on a limited number of simulations, - the study focuses on a specific type of vehicle-pedestrian collision	- Rapid impact detection and timely deployment of the active bonnet system. - Optimization of sensing and actuation subsystems for real-world active bonnet implementations. - Reproducing the detailed behaviour of pyrotechnic deployment systems in the actuator model.	- The effectiveness of active bonnet systems depends on rapid impact detection and timely deployment. - There is a need for a comprehensive analysis of pedestrian impact dynamics and the design performance of active bonnet systems.	- investigating the effect of different actuator types on the performance of active bonnet systems, - analyzing the impact of various pedestrian and vehicle parameters on the effectiveness of the system, - developing more advanced models of pedestrian and vehicle interactions	- the active bonnet system can be used to reduce pedestrian head injury severity in vehicle-pedestrian collisions, - the system's effectiveness depends on the synchronization between pedestrian detection and bonnet deployment, - optimizing the sensing and actuation subsystems is crucial for real-world implementations	- vehicle safety systems, - pedestrian protection systems, - accident reconstruction and analysis	- Numerous accident reconstruction studies have highlighted the importance of head impact with the vehicle front structure in determining injury severity. - Multibody modelling is widely used for analysing pedestrian impacts due to its ability to reproduce global kinematics of the human body. - Prior studies have employed simplified actuator representations, but the present study provides a more detailed analysis of the influence of actuator torque on head injury mitigation.	- Pedestrian head injuries are a major consequence of vehicle-pedestrian collisions. - Active bonnet systems can mitigate injury severity by increasing clearance between the hood and rigid engine components. - The effectiveness of active bonnet systems depends on rapid impact detection and timely deployment. - This study proposes a contact-triggered active bonnet system and evaluates its design performance using multibody modelling and experimental validation.	- Pedestrian safety is a major challenge in road traffic safety worldwide. - Head injuries are a critical outcome of pedestrian collisions, often leading to severe trauma or fatality. - The dynamics of pedestrian impacts involve a sequence of events, including initial contact and secondary impact. - Numerous accident reconstruction studies have highlighted the importance of head impact with the vehicle front structure in determining injury severity.	This study proposes a contact-triggered active bonnet system to mitigate pedestrian head injuries in vehicle-pedestrian collisions. The system is analyzed using MATLAB/Simulink and Simscape Multibody, and experimentally validated. The results show that the active bonnet system can reduce peak head acceleration and HIC15 values under several investigated impact conditions.
Machine Learning-Based Framework for Pre-Impact Same-Level Fall and Fall-from-Height Detection in Construction Sites Using a Single Wearable Inertial Measurement Unit ↗ os15090618 full text	2025	Biosensors	- Establish a machine learning-based approach for accurate identification of SLF, FFH, and non-fall events. - Develop a system that can detect falls in real-time and provide sufficient lead time for wearable airbag actuation.	Same-level-falls and falls-from-height remain major causes of severe injuries and fatalities on construction sites, and there is a need for accurate detection of these falls to prevent false positives.	- A machine learning-based approach using a single waist-mounted inertial measurement unit. - A two-stage feature extraction method. - An ensemble SHAP-PFI feature selector. - XGBoost, CatBoost, and LightGBM algorithms for classification.	- Type of fall (SLF or FFH), - Non-fall activities, - IMU sensor data	- Macro F1-score, - Macro Matthews correlation coefficient, - Latency, - Average lead times	- 48 participants, - 39 non-fall activities, - 10 types of SLFs, - 8 types of FFHs	- 48 participants, - Participants executed 39 non-fall activities, 10 types of SLFs, and 8 types of FFHs	- Average macro F1-score of 0.901, - Macro Matthews correlation coefficient of 0.869, - Latency of 1.51 × 10−3 ms per window, - Average lead times of 402 ms for SLFs and 640 ms for FFHs	- Falls account for roughly half of serious injuries and nearly one-third of fatalities, - 59.61% of all occupational accidents within the construction industry are falls, - 43.5% of fatal occupational incidents in the USA are falls, - The weighted XGBoost classifier achieved a macro accuracy of 0.985	- A single-sensor, waist-worn IMU integrated device can identify pre-impact SLF and FFH events, - The employment of a two-step feature extractor and an ensemble SHAP-PFI feature selector yields distinct class separation, - The XGBoost algorithm achieved the optimal compromise between accuracy, robustness, and computational efficiency	- A single-sensor, waist-worn IMU integrated device for pre-impact SLF and FFH event detection. - A two-step feature extractor for reducing initial descriptors to distinctive and informative features. - An ensemble SHAP-PFI feature selector for selecting the most relevant and influential features.	- the dataset does not encompass the full range of postures, tool usage, or surface conditions present on construction sites, - some relevant movements may have been omitted, - the dataset demonstrates a pronounced class imbalance favoring non-fall activities	- High-frequency noise in the IMU signals. - False positives due to high-vibration activities and jumps. - Achieving sufficient lead time for wearable airbag actuation.	- Current fall detection systems are prone to false positives. - There is a need for accurate and real-time fall detection systems. - Prior research has not achieved sufficient lead time for wearable airbag actuation.	—	- The device can be used as a central component for wearable fall-prevention devices in construction scenarios, - The average lead times of 402 ms for SLFs and 640 ms for FFHs surpass the 130 ms inflation time required for wearable airbags, - The device can reduce the rate of false positives in fall detection	- Construction sites, - Wearable fall-prevention devices, - Occupational safety	- Prior studies have reported strong pre-impact fall detection performance using a single waist-mounted IMU. - Other body locations have been explored for fall detection, but the waist is the most effective location. - SHAP-Select and permutation feature importance methods have been used for feature selection in other domains.	- Same-level-falls and falls-from-height are major causes of injuries and fatalities on construction sites. - Accurate detection of these falls is crucial for fall-prevention systems. - Current systems are prone to false positives. - This study proposes a machine learning-based approach for accurate identification of same-level falls, falls-from-height, and non-fall events. - The approach uses a single waist-mounted inertial measurement unit and achieves high detection accuracy and real-time processing.	- Falls are the primary cause of serious injuries and deaths in the construction industry. - Same-level falls and falls-from-height are the two main categories of falls. - Prior research has reported a persistent increase in fall-related fatalities in recent years. - There is a need for accurate and real-time fall detection systems to prevent injuries and fatalities.	This study proposes a machine learning-based approach for pre-impact same-level fall and fall-from-height detection in construction sites using a single wearable inertial measurement unit. The core technique involves a two-stage feature extraction and an ensemble SHAP-PFI method for feature selection. The main outcome is an average macro F1-score of 0.901 and macro Matthews correlation coefficient of 0.869.
Wrist-Based Fall Detection: Towards Generalization across Datasets ↗ s24051679 full text	2024	Sensors	- The proposal of this research work is the creation of a new dataset that encompasses data from three different datasets. - The model aims to estimate the classification function that distinguishes between activities of daily living and falls. - The objective is to improve the generalization properties of the model across different datasets.	The central research question is how to improve the generalization of fall detection models across different datasets.	- The creation of a new dataset that combines data from three different datasets. - The use of classic cost-sensitive machine learning methods to deal with class imbalance. - The extraction of temporal and frequency features from raw data using a sliding window of 2s with an overlap of 50%. - The evaluation of the model's performance using recall, specificity, and AUC-ROC metrics.	- type of sensor, - type of algorithm, - preprocessing process, - sampling rate	- recall, - specificity, - Area Under the Receiver Operating Characteristic curve (AUC-ROC), - accuracy, - F1-Score	- UP Fall dataset, - UMA Fall dataset, - a new dataset with more than 1300 fall samples and 28 K negative samples	- 22 participants in 12 activities, - older adults, - people who have already experienced a fall	- recall of 90.57%, - specificity of 96.91%, - an Area Under the Receiver Operating Characteristic curve (AUC-ROC) value of 98.85%, - accuracy of 91.1%, - AUC-ROC compared to the other datasets	- the UP Fall dataset demonstrates the best generalization ability to fall activities, - the UMA Fall dataset stands out in generalizing falls when used by the ANN algorithm, - the algorithms trained with the UMA Fall dataset achieved a higher AUC-ROC, - the performance of the current model does not fully meet realistic scenarios, - the values obtained for F1-Score and AUC-ROC indicate the importance of algorithm choice and preprocessing process	- the creation of a new dataset that encompasses data from three different datasets improves generalization, - ML models can be effective in detecting falls, - the choice of algorithm and preprocessing process is critical for model performance, - wrist-based fall detection can be a practical solution	- A new dataset that combines data from three different datasets to improve generalization across datasets. - A standard way of adding samples to the dataset, allowing for the future addition of other data sources. - The use of classic cost-sensitive machine learning methods to deal with class imbalance. - The evaluation of the model's performance using recall, specificity, and AUC-ROC metrics.	- small datasets, - lack of diversity in fall samples, - simulated data sets, - risks and costs associated with simulating falls, - high level of computational resources required for analysis	- The difficulty in obtaining real-life datasets of falls due to the risks and costs associated with data acquisition. - The challenge of simulating falls in a controlled environment, which requires specialized equipment and human supervision. - The need to address class imbalance in the dataset, which can affect the performance of the model.	- The lack of generalization across datasets in previous fall detection studies. - The limited number of samples in each class, which affects the performance of fall detection systems. - The need for a standardized way of adding samples to datasets to improve generalization.	- analyzing diverse datasets, - understanding the realistic limits of fall detection, - improving the generalization capacity of the model, - exploring the use of other sensors and devices	- wrist-based fall detection can be used in real-life situations, - the use of accelerometers and gyroscopes can be advantageous for detecting falls, - low energy consumption sensors can be integrated into wearable devices	- healthcare, - elderly care, - fall detection systems, - wearable devices	- Previous studies have developed fall detectors using various sensors, including accelerometers and gyroscopes. - Machine learning algorithms such as Support Vector Machines, Artificial Neural Networks, and k-Nearest Neighbors have been used for fall detection. - The use of threshold-based methods has also been explored, but machine learning approaches have shown better performance. - The importance of dataset diversity and size has been highlighted in previous studies.	- Increasing age is related to a decrease in independence of movement and a higher risk of falls. - Falls have a significant impact on the health and independence of older adults, as well as financial implications for health systems. - Previous studies have developed fall detectors using various sensors, but these models are often trained and tested on small datasets, leading to overoptimistic results. - This paper proposes the creation of a new dataset that combines data from three different datasets to improve generalization across datasets. - The new dataset includes a standard way of adding samples, allowing for the future addition of other data sources. - The model is evaluated using classic cost-sensitive machine learning methods to deal with class imbalance.	- Aging is a natural human process that leads to a decrease in independence of movement and an increased risk of falls. - The proportion of the world's population over 60 is expected to almost double from 12% to 22% between 2015 and 2050. - Previous studies have developed fall detectors using various sensors, but these models are often trained and tested on small datasets. - The lack of diversity and limited number of samples in each class affect the performance of fall detection systems. - This paper aims to address the gap in generalization across datasets.	This paper studies wrist-based fall detection, proposing a new dataset that combines data from three different datasets to improve generalization across datasets. The core technique involves using classic cost-sensitive machine learning methods to deal with class imbalance. The main outcome is a model that achieves a recall of 90.57%, a specificity of 96.91%, and an AUC-ROC value of 98.85%.
Wearable Fall Detection System with Real-Time Localization and Notification Capabilities ↗ s25123632 full text	2025	Sensors	- To develop a portable, wearable device that can detect fall events and transmit alerts in real-time. - To design a low-complexity algorithm that can meet the requirements for long-term outdoor use. - To integrate the device with a messaging application to send GPS location information and fall alert notifications.	The lack of location awareness and high complexity of many existing fall detection algorithms pose a challenge for their implementation in real-world applications, particularly in outdoor scenarios where accurate determination of the fall location is crucial for a timely emergency response.	—	- sensor placement, - wearing methods, - power consumption modes	- fall events, - GPS coordinates, - accelerometer and gyroscope signals	- a total of 6750 samples collected from fifteen test participants	- fifteen test participants, - elderly population aged 65 and above	- 6750 samples were collected, - 6000 daily activity samples, - 750 fall events, - a sampling frequency of 100Hz, - an estimated average current of 5 mA	- the proposed algorithm can filter out eight types of daily activities, - the algorithm can detect four types of falls: forward, backward, left, and right, - placing the sensor on the waist achieved an accuracy of up to 98.42%, - the proportion of elderly people in Taiwan will increase significantly to 43.6% by 2070	- the proposed system is effective in outdoor environments, - the system can detect fall events and transmit GPS coordinates, - the system meets the requirements for long-term outdoor use	- A portable, wearable device that integrates a microcontroller, an inertial sensor, and a chip module featuring GPS and NB-IoT technologies. - A low-complexity algorithm based on a finite-state machine that can detect fall events and filter out daily activities. - An integration with a messaging application to send GPS location information and fall alert notifications.	- the complexity of many existing algorithms poses a challenge for deployment on edge devices, - limited computational resources and battery life, - the study tested sensor placements on a limited number of participants	- The complexity of many existing fall detection algorithms, which poses a challenge for their implementation in real-world applications. - The lack of location awareness in many existing fall detection algorithms, which is crucial for a timely emergency response. - The need for a low-complexity and portable wearable fall detection system that can detect fall events and transmit alerts in real-time.	- The lack of location awareness in many existing fall detection algorithms. - The high complexity of many existing fall detection algorithms, which poses a challenge for their implementation in real-world applications. - The need for a portable, wearable, and low-complexity fall detection system that can detect fall events and transmit alerts in real-time.	- further reduction in power consumption, - utilizing the IMU's Wake-on-Shake feature, - investigating the impact of sensor wearing methods and placement on fall detection performance	- ensuring the safety and care of the elderly, - providing timely emergency response, - integrating the system with instant messaging applications	- wearable devices, - Internet of Things (IoT), - elderly care	- Many existing fall detection algorithms rely on inertial measurement units (IMUs) to assess the user's posture and activity. - Some studies have proposed fall detection systems based on wearable glasses or other devices, but these systems often have limitations in terms of location awareness and complexity. - The paper situates itself against prior work by proposing a wearable fall detection system that addresses the issues of location awareness and complexity.	- The paper addresses the issue of fall detection in elderly populations, which is a significant concern due to the rising proportion of elderly people. - Many existing fall detection algorithms are difficult to implement in real-world applications due to their complexity and lack of location awareness. - The proposed system integrates a microcontroller, an inertial sensor, and a chip module featuring GPS and NB-IoT technologies to detect fall events and transmit alerts. - The system is designed for long-term outdoor use and can filter out daily activities while detecting different types of falls.	- The proportion of elderly populations is rising, and ensuring their safety and care is becoming an increasingly important issue. - Falls are a leading cause of injury-related deaths among the elderly, and many existing fall detection algorithms are impractical for long-term, continuous use in practical settings. - The paper proposes a wearable fall detection system that addresses the issues of location awareness and complexity. - The system is designed to be portable, wearable, and capable of detecting fall events and transmitting alerts in real-time.	This paper studies a wearable fall detection system with real-time localization and notification capabilities, using a low-complexity algorithm based on a finite-state machine, and achieves effective fall detection in outdoor environments. The system integrates a microcontroller, an inertial sensor, and a chip module featuring GPS and NB-IoT technologies. Experimental tests demonstrated the system's effectiveness in outdoor environments.

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Experimental Study of Long Short-Term Memory and Transformer Models for Fall Detection on Smartwatches ↗ s24196235 full text	2024	Sensors	- To investigate and experiment with an LSTM model for fall detection on a smartwatch. - To compare the performance of three variants of LSTM and two variants of Transformer models for learning fall patterns. - To evaluate the real-time performance of the best model trained with SmartFallMM data using the SmartFall App.	The system misses some falls and generates an annoying amount of False Positives for practical use in real-time fall detection applications on smartwatches.	- The use of three variants of LSTM and two variants of Transformer models for fall detection. - The employment of a Leave-One-Out strategy for training and evaluation. - The use of the SmartFall App for real-time testing of the models. - The collection of data from three datasets: UniMib, Kfall, and SmartFallMM.	- Type of model (LSTM, Transformer, CNN-LSTM), - Hyperparameters (e.g. window size, batch size), - Type of dataset (UniMib, K-fall, SmartFallMM)	- Fall detection accuracy, - Recall, - False Positives	- UniMib dataset with 30 subjects, - K-fall dataset, - SmartFallMM dataset	- 30 subjects with ages ranging from 18 to 60 years in the UniMib dataset, - Subjects in the K-fall and SmartFallMM datasets	- The CNN-LSTM model was better than the Transformer model for all the datasets in offline training, - The Transformer is a preferable choice for deployment in real-time fall detection applications, - A threshold algorithm missed 23% of falls in the real-world test, - A deep learning-based model has almost 100% Recall, - The real-time evaluation confirmed that the Transformer outperforms LSTM in dynamic real-world scenarios, - The Recall of a model using a threshold-based algorithm was 77%	- The CNN-LSTM model was better than the Transformer model for all the datasets, - The Transformer outperforms LSTM in dynamic real-world scenarios, - A threshold algorithm missed 23% of falls in the real-world test, - A deep learning-based model has almost 100% Recall, - The Recall of a model using a threshold-based algorithm was 77%, - The UniMib dataset has 30 subjects, with ages ranging from 18 to 60 years	- The Transformer outperforms LSTM in dynamic real-world scenarios, - A deep learning-based model has almost 100% Recall, - The real-time evaluation confirmed that the Transformer is a preferable choice for deployment in real-time fall detection applications, - The system is not restricted to the Huawei brand of watch, - Any smartwatch that runs WearOS can be used	- A comparative study of LSTM and Transformer models for fall detection on smartwatches. - The development of a real-time fall detection system using the SmartFall App. - An evaluation of the performance of different variants of LSTM and Transformer models for fall detection.	- Falls are rare events, and it is very labor-intensive to collect large amounts of fall data, - The dataset is not very large, - Some differences in hyperparameters were unavoidable due to the salient differences in architecture	- The rarity of fall events, making it labor-intensive to collect large amounts of fall data. - The need for a system that can accurately detect falls in real-time with minimal False Positives. - The limitation of wearable devices in terms of power and computation constraints.	- The lack of successful real-time fall detection applications on commodity-based smartwatches. - The need for a comparative study of different machine learning models for fall detection. - The requirement for a system that can accurately detect falls in real-time with minimal False Positives.	- Comparative studies involving real-time testing of deep learning-based fall detection models on a smartwatch, - Real-world evaluation studies using different smartwatches, - Investigation of other deep learning models for fall detection	- The use of deep learning-based models for fall detection can reduce the number of missed falls, - The use of the Transformer model can improve the performance of fall detection systems in real-time scenarios, - Any smartwatch that runs WearOS can be used for fall detection	- Fall detection on smartwatches, - Human activity recognition, - Real-time fall detection applications	- Prior work has developed wearable devices with fall detection systems using various machine learning models. - The Transformer model has been shown to achieve high accuracy for human activity recognition. - The LSTM model has been used for fall detection, but its performance in real-time applications is limited.	- Falls are the second leading cause of unintentional injury deaths worldwide. - Adults older than 60 suffer the greatest number of fatal falls. - Wearable devices with fall detection systems have been developed, but none have been successfully used in real-time applications on commodity-based smartwatches. - This paper compares the performance of LSTM and Transformer models for fall detection on smartwatches.	- Falls are a significant cause of injury and death, particularly among older adults. - Prior work has developed wearable devices with fall detection systems, but these have limitations. - There is no consensus on the best architecture for fall detection using accelerometer data from a wristwatch. - This paper aims to address the gap in real-time fall detection on smartwatches.	This paper studies the use of Long Short-Term Memory and Transformer models for fall detection on smartwatches, with a focus on real-time performance. The core technique involves comparing the performance of different variants of these models. The main outcome is that the Transformer model outperforms LSTM in dynamic real-world scenarios.
Analyzing Optimal Wearable Motion Sensor Placement for Accurate Classification of Fall Directions ↗ s24196432 full text	2024	Sensors	- This study aims to identify the optimal placement of wearable sensors for effective fall-direction classification. - The research was conducted in three phases: comparing classifiers, evaluating performance differences between sensors on the left and right sides, and exploring the efficacy of combining sensors from upper and lower body regions.	The central research question is to determine the optimal placement of wearable sensors for accurate classification of fall directions.	- The study uses inertial measurement unit sensors at 12 distinct body locations. - The research employs supervised learning models, including support vector machine, random forest, k-nearest neighbors, and multilayer perceptron. - The study uses a 1D Gaussian filter for data preprocessing and stratified 10-fold cross-validation for evaluation.	- sensor placement on the body, - type of classifier used (e.g. SVM, RF, KNN, MLP)	- accuracy, - precision, - recall (sensitivity), - F-score	- a dataset with four categories: non-fall, backward fall, forward fall, and lateral (side) fall	- participants were asked to fall 1-10 times in each direction, - the number of participants is not specified	- up to 600 samples recorded at 60 Hz, - 10 s of data per sequence, - exceptionally high accuracy and F-score, - Recall (sensitivity) = TP / (TP + FN), - precision, recall (sensitivity), and F-score were used as evaluation metrics	- the support vector machine (SVM) is more effective than other classifiers across all sensor locations, - sensors positioned on the pelvis and upper legs in the lower body were the most effective, - sensors positioned on the shoulder and head in the upper body were the most effective, - no significant performance differences within the same anatomical areas between the left and right sensor locations	- combining sensors across both upper and lower body parts is more effective for accurate fall-direction classification, - single-sensor placements are often sufficient for general fall detection but inadequate for accurate fall-direction classification, - the SVM classifier emerged as the best classifier for fall-direction classification	- The study identifies the most effective sensor locations for fall-direction classification, including the pelvis, upper legs, shoulder, and head. - The research develops a methodology for selecting specific sensor locations and segmenting body parts. - The study employs supervised learning models to assess sensor locations and determine the most effective sensor placements for each body part.	- sensor placement limitations or challenges in detecting lateral falls using the current setup	- The study faces the challenge of identifying the most effective sensor locations for fall-direction classification. - The research must address the issue of sensor placement on the body and its impact on classification accuracy. - The study must also consider user comfort and practical placement considerations when selecting sensor locations.	- Limited attention has been given to the classification of fall direction across different body regions. - Previous studies have primarily focused on distinguishing falls from non-falls, rather than fall-direction classification.	- explore the performance of alternative classifiers specifically for fall-direction classification - explore optimal combinations of upper and lower body sensors to maximize accuracy while minimizing the number of sensors required - investigate into optimizing sensor configurations or applying more advanced machine learning models to improve the detection of lateral falls - explore this aspect to better replicate real-world falls and improve the robustness of fall-direction classification systems	- wearable sensor systems can enhance the well-being of patients and older adults by assessing fall risk, detecting falls, and monitoring other health-related factors, - accurate fall-direction classification can lead to better prevention and intervention strategies, - sensor placement on the body can significantly influence the accuracy and efficiency of classification or detection systems	- fall detection and prevention, - health monitoring, - rehabilitation	- Previous research has identified optimal sensor locations for distinguishing falls from non-falls. - Studies have highlighted the waist, chest, and thigh as key sensor locations for fall detection. - The paper situates itself against prior work by focusing on fall-direction classification rather than fall detection.	- Falls are a significant risk factor, necessitating accurate classification methods. - This study aims to identify the optimal placement of wearable sensors for effective fall-direction classification. - Previous research has focused on distinguishing falls from non-falls, but limited attention has been given to fall-direction classification. - The study assesses inertial measurement unit sensors at 12 distinct body locations to determine the most effective positions for capturing fall-related data.	- Falls are a major public health concern, ranking as the second leading cause of injuries and accidental death globally. - Wearable sensor systems enhance the well-being of patients and older adults by assessing fall risk, detecting falls, and monitoring other health-related factors. - Previous studies have primarily focused on identifying optimal sensor locations for distinguishing between fall events and non-fall activities. - The accuracy and efficiency of classification or detection systems are significantly influenced by sensor placement on the body.	This study analyzes the optimal placement of wearable sensors for accurate classification of fall directions, using inertial measurement unit sensors and machine learning algorithms, and finds that sensors on the pelvis, upper legs, shoulder, and head are most effective. The research aims to identify the most effective sensor locations for fall-direction classification. The study uses a support vector machine classifier and achieves high accuracy.
A Low-Cost Inertial Measurement Unit Motion Capture System for Operation Posture Collection and Recognition ↗ s24020686 full text	2024	Sensors	- The paper aims to propose a low-cost human motion capture system based on BMI 160. - The objective is to recognize human posture in factories for human-machine collaboration and workflow improvement.	The central research question is how to develop a low-cost and effective solution for human posture recognition in factories.	- BMI 160, a type of six-axis inertial measurement unit. - WIFI communication to process collected data. - Kalman filtering algorithm for filtering and drift compensation of IMU signals. - Bi-directional long short-term memory (BiLSTM) model for posture recognition.	- type of inertial measurement unit (IMU), BMI 160, - number and placement of inertial sensors, six sensors on the chest, waist, knee joints, and ankle joints of both legs	- human posture, - rotation angles around XYZ directions, - displacement in XYZ directions	- 16,067 labeled samples, - simulated assembly operations in a virtual reality environment	- a single participant with a mechanical engineering background, - the participant wore six inertial sensors on the chest, waist, knee joints, and ankle joints of both legs	- 16,067 labeled samples were obtained for posture recognition model training, - the accumulated displacement and the rotation angle of six joints in the three directions were used as input features, - the bi-directional long short-term memory (BiLSTM) model was used to identify seven common operation postures, - an average accuracy of 98.24% was achieved	- the average accuracy of BiLSTM is 98.24%, - the average accuracy of LSTM is 95.81%, - BiLSTM has a greater advantage in basic posture classification for assembly tasks, - BiLSTM requires longer training time, 75.08 s, - the difference in prediction time between the two models is not significant, 15ms and 31 ms	- the proposed method could be used to develop a low-cost and effective solution to human posture recognition for factory operation, - integrating more features and employing more complex machine learning models may result in higher recognition accuracy, - the proposed system is suitable for factories due to its portability and resistance to obstruction	- A low-cost human motion capture system based on BMI 160. - A digital human model established on Unity3D to synchronously visualize and present human movements. - A bi-directional long short-term memory (BiLSTM) model to identify seven common operation postures.	- the experiment involved a single participant with a mechanical engineering background, - the participant wore six inertial sensors on the chest, waist, knee joints, and ankle joints of both legs, - the wearable device required calibration every 10 minutes due to drift	- Drift of the BMI160, causing body parts to face the wrong direction after some time. - Frequency congestion in the experimental environment, causing excessive data transmission delay. - Ensuring low data transmission delay in the experimental environment.	- Existing inertial sensing solutions are generally expensive. - There is a need for a low-cost and effective solution for human posture recognition in factories.	- integrating more features and employing more complex machine learning models may result in higher recognition accuracy, - comparing the recognition accuracy of different machine learning models	- the proposed system can facilitate human-machine collaboration, - the system can contribute to human factor analysis and workflow improvement, - the system can be used to recognize human body movement in various scenarios	- factory operation, - human-machine collaboration, - digital human modeling, - assembly and maintenance operation processes of large-scale equipment	- Prior work has established methods for capturing full-body motion using sparse inertial sensors. - Susperregi et al. proposed the fusion of multiple low-cost sensors and cameras to capture human behavior. - Caputo et al. utilized a motion capture system to estimate the basic segment positions of the human body.	- Human posture recognition is important for human-machine collaboration and workflow improvement in factories. - Inertial sensors have advantages over optical sensors in terms of portability and resistance to obstruction. - Existing inertial sensing solutions are generally expensive. - This paper proposes a low-cost human motion capture system based on BMI 160. - The system achieves an average accuracy of 98.24% in recognizing seven common operation postures.	- Human posture recognition is important for human-machine collaboration and workflow improvement in factories. - Inertial sensors have advantages over optical sensors in terms of portability and resistance to obstruction. - Prior work has established methods for capturing full-body motion using sparse inertial sensors. - The gap is that existing inertial sensing solutions are generally expensive.	This paper studies human posture recognition in factories using a low-cost inertial measurement unit motion capture system. The core technique is based on BMI 160, a type of six-axis inertial measurement unit. The main outcome is an average accuracy of 98.24% in recognizing seven common operation postures.
Measuring the Effect of an Ergonomic Lecture on the Rapid Upper Limb Assessment Scores of Dental Assistant Students Using Inertial Sensor-Based Motion Capture—A Randomized Controlled Study ↗ re12161670 full text	2024	Healthcare	- Educating dental assistant students on ergonomic principles for use in the workplace would lead to a significant improvement in their working posture.	The central research question is how an ergonomic lecture and training intervention affects postural risk and musculoskeletal disorders in dental assistant students.	- The study used a randomized controlled trial (RCT) design with a parallel-group and pre-post-test design. - Inertial sensor-based motion capture was used to assess postural risk. - The Rapid Upper Limb Assessment (RULA) score was used to evaluate posture, muscle engagement, and external loads.	- Ergonomic lecture - Five-week strengthening and stretching training routine - Intervention group (IG) vs control group (CG)	- Rapid Upper Limb Assessment (RULA) scores - Part-scores of the upper arm and wrist - MSD prevalence	- Nordic Questionnaire on MSD prevalence - 90 s MoCap with Xsens IMU	- 18 female dental assistant students - Age: 19.44 ± 6.83 years - Height: 164.59 ± 5.32 cm - Weight: 64.88 ± 16.52 kg - BMI: 19.70 ± 4.89 kg/m2	- Mixed ANOVA results showed no significant interaction between groups and time of measurements (Greenhouse-Geisser: F(1,6) = 0.226; p = 0.651; ηp 2 = 0.036) - There was a significant reduction between the time of measurements (F(1,6) = 34.940; p = 0.001; ηp 2 = 0.853) - Bonferroni-adjusted post hoc analysis revealed a significant (p = 0.001) score reduction between the time of measurement from preto post-test (Mean Diff = -0.560; 95% confidence interval [CI], -0.792 to -0.328) - The participants completed the Nordic Questionnaire on MSD prevalence - A 90 s MoCap with Xsens IMU was performed	- 98.8% lifetime MSD prevalence rate among dentists and dental students - 92% 12-month MSD prevalence rate among dentists and dental students - 65.6% 7-day MSD prevalence rate among dentists and dental students - Significant difference in the Rapid Upper Limb Assessment (RULA) and part-scores of the upper arm and wrist - No significant differences in the time of measurement and groups were detected after the five-week training intervention and the 18-week follow-up questionnaire	- A targeted ergonomics lecture was effective for dental assistant students - Technologies such as IMU improved workplace ergonomics in dentists - Educating dental assistant students on ergonomic principles for use in the workplace would lead to a significant improvement in their working posture - The study showed a significant reduction in MSDs	- The study evaluates the effects of an ergonomic lecture and training intervention on postural risk and MSDs in dental assistant students. - The study uses inertial sensor-based motion capture to assess the effects of the intervention. - The study provides insights into the effectiveness of a targeted ergonomics lecture for dental assistant students.	- Small sample size - Limited measurement period - No significant differences in the time of measurement and groups were detected after the five-week training intervention and the 18-week follow-up questionnaire - Further studies with a longer measurement period, follow-up, and larger sample sizes are recommended	- Musculoskeletal disorders (MSDs) are a significant problem among dental professionals due to monotonous and one-sided physical exertion. - The study had to account for the potential impact of the intervention on the participants' working posture. - The study had to ensure the accuracy and reliability of the inertial sensor-based motion capture data.	- There is a need for effective interventions to mitigate musculoskeletal disorders (MSDs) among dental professionals. - Prior studies have reported a high prevalence of MSDs among dentists and dental students.	- Further studies with a longer measurement period, follow-up, and larger sample sizes are recommended - Investigation of the effects of ergonomic lectures and training interventions on MSDs in other professions	- Ergonomic lectures can be effective in reducing MSDs among dental assistant students - The use of IMU-based motion capture can improve workplace ergonomics - Dental assistant students should be educated on ergonomic principles for use in the workplace	- Dentistry - Dental education - Occupational health and safety - Ergonomics	- A study included 540 dentists and dental students and reported lifetime, 12-month, and 7-day MSD prevalence rates of 98.8%, 92%, and 65.6%, respectively. - Hayes et al. found a prevalence of musculoskeletal pain ranging from 64% to 93%. - German prevalence rates of MSDs mirror international findings.	- Individuals working in dentistry have a high prevalence of musculoskeletal disorders (MSDs) due to monotonous and one-sided physical exertion. - Inertial measurement units (IMU) are used to assess postural risk at work. - The study aims to evaluate the effects of an ergonomic lecture and training intervention on postural risk and MSDs in dental assistant students. - The study uses inertial sensor-based motion capture to assess the effects of the intervention.	- Day-to-day work in dentistry requires physical exertion, making the musculoskeletal system susceptible to injury. - The occurrence of musculoskeletal disorders (MSDs) among dentistry employees is notably more frequent than in the general population. - A study found a high prevalence of MSDs among dentists and dental students. - The study aims to address the need for effective interventions to mitigate MSDs among dental professionals.	This study evaluates the effects of an ergonomic lecture and training intervention on postural risk and musculoskeletal disorders in dental assistant students using inertial sensor-based motion capture. The core technique used is the Rapid Upper Limb Assessment (RULA) score. The main outcome is a significant reduction in RULA scores, indicating improved working posture.
Bi-LSTM Network for Multimodal Continuous Human Activity Recognition and Fall Detection ↗ li2019bilstm full text + collection	2019	IEEE Sensors Journal	- To propose a framework for human activity recognition and fall detection using multi-layer bi-LSTM network and multimodal sensor fusion. - To improve classification performance and reduce accuracy variance.	How to accurately recognize human activities and detect falls using multimodal sensor fusion?	- Multi-layer bi-LSTM network. - Multimodal sensor fusion using FMCW radar and wearable inertial sensors. - Soft and hard fusion methods. - Sequential Backward Selection for feature selection.	- Type of sensor (FMCW radar, wearable inertial sensors). - Number of participants.	- Classification accuracy. - Fall detection rate.	- The dataset consists of continuous activity streams from FMCW radar and three wearable inertial sensors.	- 15 male and 1 female participants aged 21-35 years.	- The proposed method achieves approximately 96% accuracy in identifying continuous activities and fall events. - The average classification accuracy increases to 94.7% using soft fusion. - The fall detection rate improves by around 10.9% using fusion.	- The proposed method achieves approximately 96% accuracy. - The average classification accuracy increases to 94.7% using soft fusion. - The fall detection rate improves by around 10.9% using fusion. - The method reduces accuracy variance among different participants by up to 18.1%. - The method increases minimum, worst-case accuracy by up to 16.2%.	- The proposed bi-LSTM network with hybrid fusion scheme achieves high accuracy in human activity recognition and fall detection. - The fusion scheme reduces accuracy variance among different participants. - The method is validated using a realistic leaving one participant out approach.	- A novel hybrid fusion scheme combining soft and hard fusion methods. - A multi-layer bi-LSTM network for multimodal sensor fusion. - Improved classification performance and reduced accuracy variance.	- The study uses a limited number of participants (15 male and 1 female). - The data collection is done in a controlled environment.	- Random transitions between activities. - Variability in participant movements and sensor data.	- Conventional methods may not capture random transitions between activities accurately. - There is a need for a novel approach that can improve classification performance and reduce accuracy variance.	- Further studies can be done to improve the accuracy and robustness of the proposed method. - The method can be applied to various applications such as healthcare and surveillance.	- The proposed method can be used for various applications such as healthcare and surveillance. - The method can improve the accuracy of human activity recognition and fall detection systems.	- Healthcare. - Surveillance. - Fall detection systems.	- Prior work uses conventional fixed-duration snapshots for activity recognition. - Some methods use sensor fusion but may not capture random transitions accurately. - This paper proposes a novel approach using multi-layer bi-LSTM network and hybrid fusion scheme.	- Human activity recognition and fall detection are important for various applications. - Conventional methods use fixed-duration snapshots and may not capture random transitions between activities. - This paper proposes a framework based on multi-layer bi-LSTM network for multimodal sensor fusion. - The framework achieves high accuracy and reduces accuracy variance.	- Human activity recognition is a challenging task due to random transitions between activities. - Conventional methods may not capture these transitions accurately. - This paper proposes a novel approach using multi-layer bi-LSTM network and multimodal sensor fusion. - The approach aims to improve classification performance and reduce accuracy variance.	This paper studies human activity recognition and fall detection using a multi-layer bi-LSTM network with multimodal sensor fusion, achieving approximately 96% accuracy. The core technique is a novel hybrid fusion scheme combining soft and hard fusion methods. The main outcome is improved classification performance and reduced accuracy variance.
Square-ring mass triaxial piezoresistive accelerometer with decoupled sensing elements for matched three-axis performance ↗ wan2026squarering full text + collection	2026	Microsystem Technologies	- A key design goal for triaxial accelerometers is to achieve consistent performance across all three axes. - To achieve matched three-axis performance, the accelerometer is designed with decoupled sensing elements. - The paper aims to provide a theoretical basis for the design of highly consistent triaxial piezoresistive accelerometers.	The different deformation mechanisms and fabrication requirements of in-plane and out-of-plane sensing structures often result in sensitivity mismatch and cross-axis coupling in triaxial accelerometers.	- The paper uses a simplified process for fabricating the prototype devices, involving first etching to expose the beam regions, then forming the piezoresistive bridge, and finally etching to release the beam-mass structure. - A Wheatstone bridge supply voltage of 5 V and an output amplification factor of approximately 20 were used to measure the accelerometer outputs. - Analytical expressions for the displacement and resonant frequencies along the three axes were derived.	- Supply voltage - Amplification factor	- Acceleration signals - Sensitivities in the x-, y-, and z-axes	—	—	- 0.41 mV/g/V - 0.40 mV/g/V - 0.54 mV/g/V - 5 V - 20× amplification	- 0.41 mV/g/V sensitivity in the x-axis - 0.40 mV/g/V sensitivity in the y-axis - 0.54 mV/g/V sensitivity in the z-axis - 20× amplification - 5 V supply voltage	- A piezoresistive three-axis accelerometer featuring a square-ring proof mass suspended from a central anchor through sensing beams is presented - The independent sensing beams and integrated piezoresistors enable decoupled readout of the three-axis acceleration signals - Analytical expressions for the displacement and resonant frequencies along the three axes were derived	- The paper presents a piezoresistive three-axis accelerometer featuring a square-ring proof mass suspended from a central anchor through sensing beams. - The independent sensing beams and integrated piezoresistors enable decoupled readout of the three-axis acceleration signals and reduce inter-axis crosstalk through structural design. - Analytical expressions for the displacement and resonant frequencies along the three axes were derived, providing a theoretical basis for the design of highly consistent triaxial piezoresistive accelerometers.	—	- The differing techniques and accuracies for in-plane and out-of-plane manufacturing pose a challenge for achieving matched performance among the three axes. - Sensitivity mismatch and cross-axis coupling are challenges in triaxial accelerometers. - Stress-induced deformation due to an asymmetric oxide layer on the top and bottom surfaces of the beam is a challenge in the conventional process for piezoresistive accelerometers.	- The techniques and accuracies for in-plane and out-of-plane manufacturing differ, making it challenging to maintain matched performance among the three axes. - Different sensing structures must be employed to detect in-plane and out-of-plane accelerations, resulting in sensitivity mismatch and cross-axis coupling.	—	- The accelerometer can be used in various applications such as medical field, automotive applications, industrial fault diagnosis, geological seismic monitoring, and defense applications - The accelerometer has a compact size, light weight, and low cost - The accelerometer enables intelligent tire pressure monitoring, vibration measurement, and instability detection	- Medical field - Automotive applications - Industrial fault diagnosis - Geological seismic monitoring - Defense applications	- Monolithically integrating three single-axis accelerometers on a single chip is one approach for three-axis accelerometers. - Adopting a single-proof-mass three-axis structure is another approach. - Prior work includes the use of MEMS technology for triaxial accelerometers, but the differing techniques and accuracies for in-plane and out-of-plane manufacturing pose a challenge.	- Micro-Electro-Mechanical Systems (MEMS) triaxial accelerometers have gained widespread adoption across diverse sectors. - They are used in various fields such as medical, automotive, industrial, and defense applications. - A key design goal for triaxial accelerometers is to achieve consistent performance across all three axes. - This paper presents a piezoresistive three-axis accelerometer featuring a square-ring proof mass suspended from a central anchor through sensing beams.	- MEMS triaxial accelerometers have gained widespread adoption across diverse sectors due to their compact size, light weight, and low cost. - They are used in various fields such as medical, automotive, industrial, and defense applications. - A key design goal for triaxial accelerometers is to achieve consistent performance across all three axes. - The techniques and accuracies for in-plane and out-of-plane manufacturing differ, making it challenging to maintain matched performance among the three axes. - This paper presents a piezoresistive three-axis accelerometer featuring a square-ring proof mass suspended from a central anchor through sensing beams.	This paper studies a square-ring mass triaxial piezoresistive accelerometer with decoupled sensing elements, using a core technique of independent sensing beams and integrated piezoresistors, and achieves a main outcome of matched three-axis performance. The accelerometer is designed to reduce inter-axis crosstalk and provide consistent performance across all three axes. The paper presents a theoretical basis for the design and fabrication of the accelerometer.

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